

ETHICAL ALIGNMENT OF SMALL LANGUAGE MODELS THROUGH KNOWLEDGE DISTILLATION FROM LLMs

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ABSTRACT

The ethical alignment of language models is a critical challenge in the development of responsible AI systems, particularly as these models are increasingly integrated into decision-making processes across domains such as healthcare, education, and public policy. While large language models (LLMs) demonstrate advanced ethical reasoning, their high computational demands limit their accessibility and deployment in low-resource environments. This research addresses this gap by enhancing the ethical reasoning capabilities of small language models (SLMs) such as LLaMA 3.2 1B, LLaMA 3.2 3B, Gemma 2 2B, and Phi 3.5 Mini 4B through fine-tuning on the ETHICS dataset and leveraging knowledge distillation from larger pre-trained models. The methodology combines structured data preprocessing, supervised fine-tuning, and advanced distillation techniques—including logit-based and response-based distillation—to transfer ethical insights from LLMs to SLMs. Models are evaluated using both quantitative metrics (e.g., accuracy and human-alignment scores) and qualitative analysis of ethical decision-making scenarios. The expected outcomes include computationally efficient SLMs with improved ethical reasoning, making them suitable for deployment in edge and mobile applications.

INTRODUCTION

In recent years, the integration of artificial intelligence into decision-making systems has raised significant ethical concerns, especially as these systems become more autonomous and widely adopted. Large language models (LLMs), such as GPT-4 and Claude, have shown considerable potential in understanding and responding to ethically complex situations. However, their high computational requirements and large memory footprints make them impractical for many real-world applications, particularly those in resource-constrained environments. On the other hand, small language models (SLMs) offer faster inference, lower power consumption, and easier deployment but often lack the ethical reasoning sophistication of their larger counterparts. This imbalance underscores a pressing need to bridge the ethical capability gap between LLMs and SLMs, ensuring that lightweight models can also make ethically sound decisions.

RQ1: How can we enhance the ethical decision-making capabilities of small language models in open-ended real-world scenarios?

RQ2: What techniques can effectively transfer ethical alignment from large language models to small, resource-efficient models?

RQ3: How can we ensure that small models deployed in the wild are resistant to manipulation and reflect human moral values?

METHODOLOGY

This work presents a novel bidirectional alignment framework for ethical reasoning in AI systems, built upon a training pipeline termed *Reverse Knowledge Distillation* (RKD). In contrast to traditional paradigms where large teacher models guide smaller ones, we reverse this flow: Small Language Models (SLMs) first learn ethical alignment and then transfer this reasoning to larger models.

Problem Formulation Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ be a dataset of natural language statements $x_i \in \mathbb{R}^p$ and binary ethical labels $y_i \in \{0, 1\}$. The objective is to first fine-tune a student model S_θ and use it to guide a teacher model T_ϕ through reverse distillation.

The student output is defined as $p_s(y | x_i; \theta) = \text{softmax}(S_\theta(x_i))$ and the teacher output as $p_t(y | x_i; \phi) = \text{softmax}(T_\phi(x_i))$. The total loss used for training is:

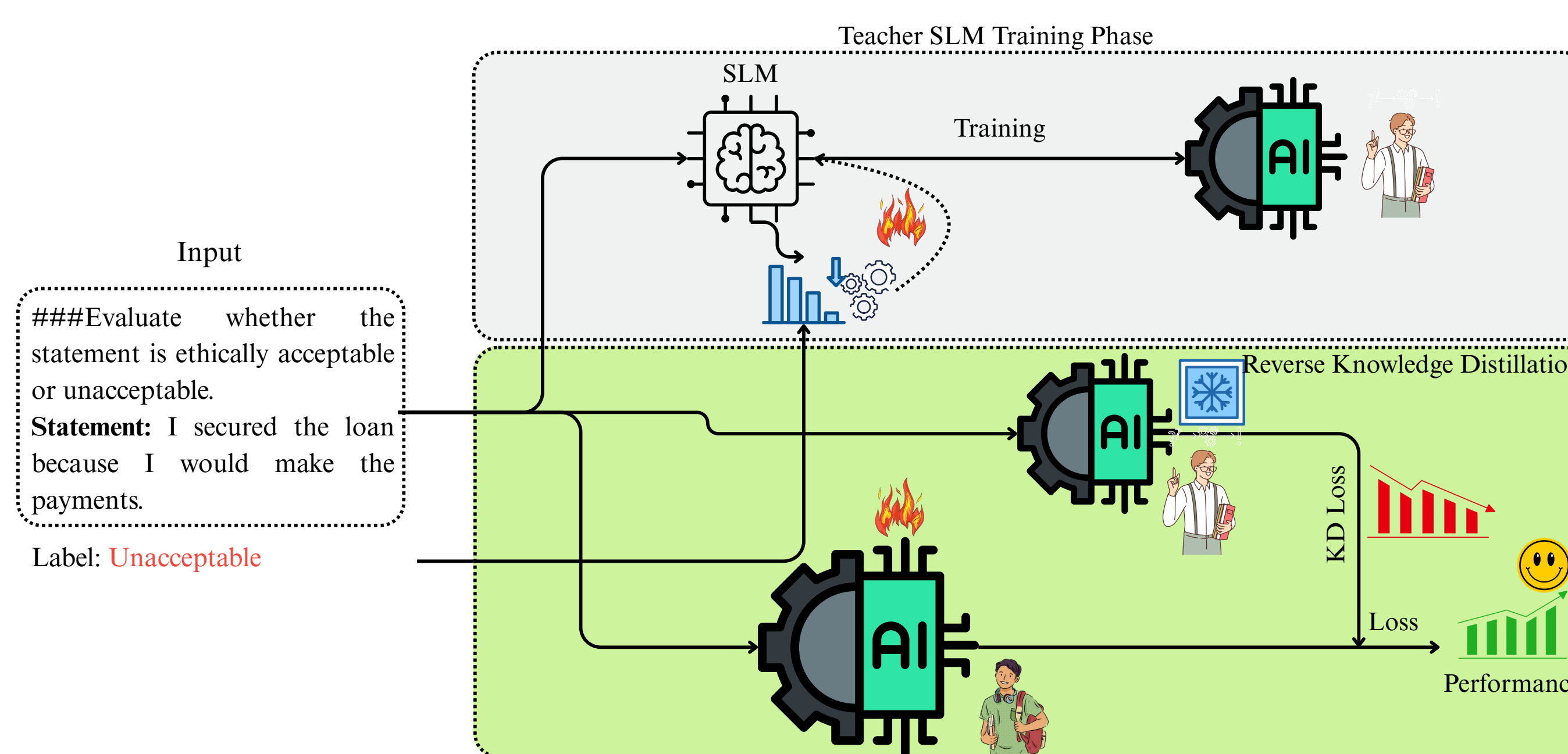
$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda \cdot \mathcal{L}_{\text{RKD}}$$

where the supervised term is:

$$\mathcal{L}_{\text{sup}} = - \sum_{i=1}^N \sum_{k=1}^K \mathbb{I}_{[y_i=k]} \log p_s(k | x_i)$$

and the distillation term:

$$\mathcal{L}_{\text{RKD}} = \sum_{i=1}^N \text{KL}(p_t(y | x_i) \| p_s(y | x_i))$$



Reverse Knowledge Distillation Framework for Ethical Reasoning with SLMs

(a) Student Supervision: Models such as LLaMA 3.2, Phi 3.5 Mini, and Gemma 2 are first fine-tuned using ethical classification objectives.

(b) Teacher Alignment: Larger models like LLaMA 2 7B and Gemma 2 9B are then aligned using soft outputs from SLMs, improving ethical reasoning in higher-capacity models.

(c) Optimization Strategy: Joint optimization over both supervised and RKD losses ensures that the student refines its performance while also shaping the ethical sensitivity of the teacher model.

RESULTS

We present a detailed analysis of the experimental outcomes, focusing on three key phases: (1) base performance of student models without fine-tuning, (2) supervised fine-tuning of student models on the ETHICS dataset, and (3) reverse knowledge distillation (RKD) applied to larger teacher models.

We evaluate ethical classification using accuracy (overall correctness) and F1 score (harmonic mean of precision and recall) to account for any class imbalance. Models are compared at three stages—Base, Fine-tuned, and Reverse KD (after distilling student improvements back into larger teachers)—to quantify both raw performance gains and their transfer to high-capacity models.

Training Hyperparameters. The following hyperparameters were used for fine-tuning LLaMA-2 (7B):

No. of training epochs	10
Train batch size	16
Learning rate	2×10^{-4}
Weight decay	0.001
Optimizer	AdamW

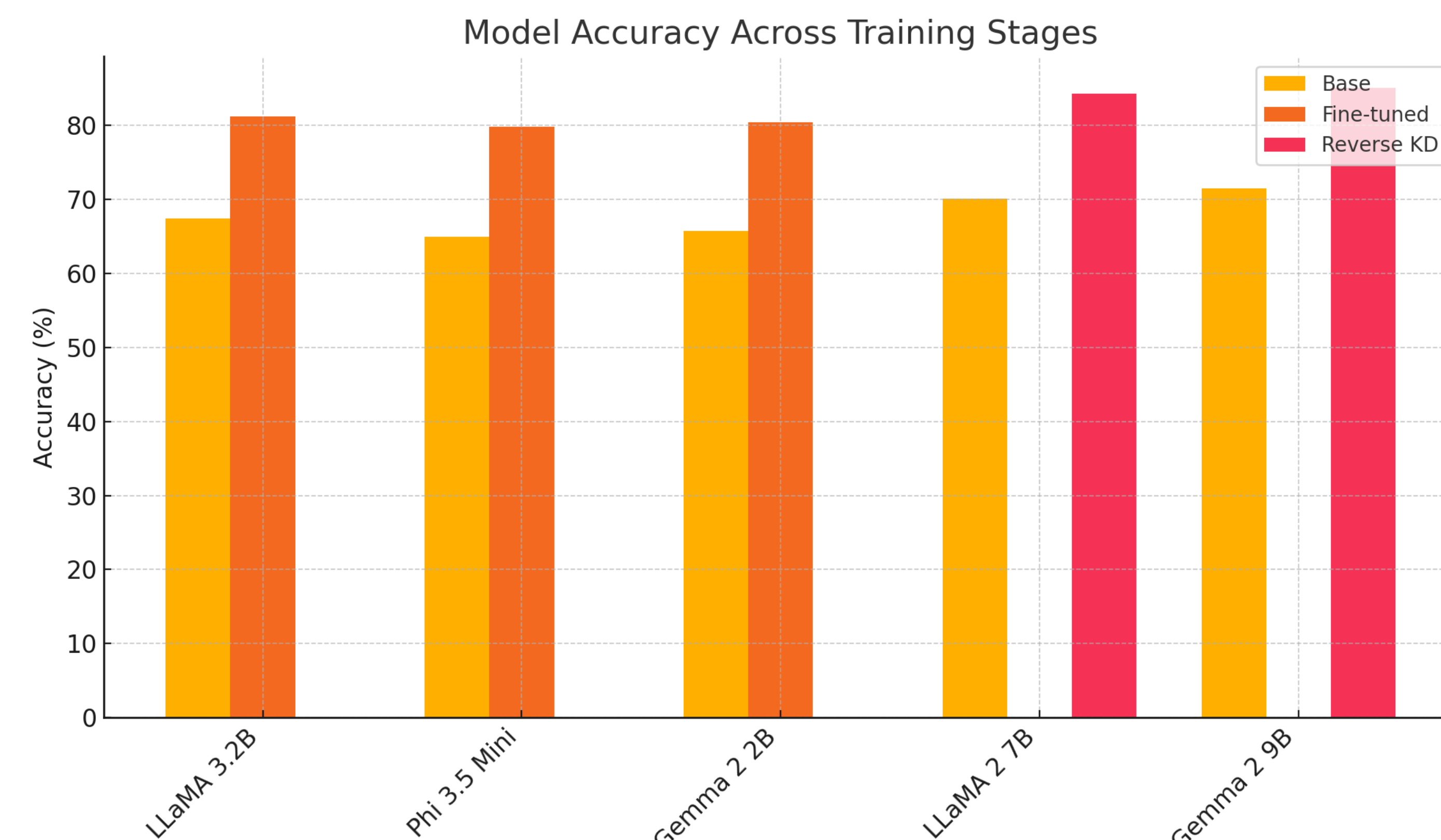
Improvements from Base → Fine-tuned → Reverse KD:

Model	Model Performance					
	Base		Fine-tuned		Reverse KD	
	Acc. (%)	F1	Acc. (%)	F1	Acc. (%)	F1
LLaMA 3.2B	67.4	0.68	81.2	0.82	—	—
Phi 3.5 Mini	64.9	0.66	79.8	0.80	—	—
Gemma 2 2B	65.7	0.67	80.4	0.81	—	—
LLaMA 2 7B	70.1	0.72	—	—	84.3	0.85
Gemma 2 9B	71.5	0.73	—	—	85.1	0.86

tableAccuracy and F1 across Base, Fine-tuned, and Reverse KD stages.

Key Observations:

- Fine-tuning boosted SLM accuracy by 12–14 points (e.g., LLaMA 3.2B from 67.4 % to 81.2 %).
- RKD enabled larger teachers to reach 84–85 % accuracy, closing the gap with fine-tuned students.
- Compact, aligned SLMs deliver strong ethical reasoning at a fraction of compute cost.



Accuracy (%) for each model at Base (gold), Fine-tuned (orange), and Reverse KD (red) stages. Fine-tuned SLMs (LLaMA 3.2 , Phi 3.5 Mini, Gemma 2)

DISCUSSION

Applying supervised fine-tuning on the ETHICS dataset significantly boosts small models' ethical classification performance (up to +14% accuracy and +0.15% F1), validating the value of direct alignment with human-annotated data. Reverse Knowledge Distillation then propagates these gains to larger teacher models, enabling them to reach >84% accuracy and >0.85% F1 without additional labeled supervision. Together, these results demonstrate a lightweight, scalable pipeline for ethical reasoning that can be deployed in resource-constrained environments. Future work will investigate multi-class ethical judgments and cross-domain generalization. We will also explore the robustness of our aligned models against adversarial and out-of-distribution inputs to ensure reliability in real-world scenarios. Additionally, integrating explainability mechanisms and human-in-the-loop feedback could further enhance user trust and facilitate safer deployments.